

5. SEMIPOSITIVE $\mathbf{R}$ -PL FUNCTIONS

Let k be a non-trivially valued non-Archimedean field. We consider a good, strictly k -analytic space X . We refer to 2.5 for the definition of PL functions. Recall from the introduction that an $\mathbf{R}$ -PL function $u: X \rightarrow \mathbf{R}$ is called *semipositive* at $x \in X$ if the associated residue line bundle $L_u(x) \in \text{Pic}(\text{red}(X, x))_{\mathbf{R}}$ is nef, where we refer to [GJR21, 6.16–6.19] for residue line bundles. Equivalently, there is a strictly affinoid neighbourhood U of x such that $u = h_{\mathcal{L}}$ for a formal k° -model $(\mathfrak{U}, \mathcal{L})$ of $(U, \mathcal{O}_U)$ with $\mathcal{L}$ nef [GJR21, Proposition 7.1].

We will show first that for an $\mathbf{R}$ -PL function on a smooth curve, semipositivity is the same as subharmonicity in the sense of Thuillier’s thesis [Thu05, Définition 3.1.5]. Recall that Thuillier defines a subharmonic function on a smooth k -analytic curve X as a semicontinuous function $u: X \rightarrow \mathbf{R} \cup \{-\infty\}$ which is not identically $-\infty$ on any connected component of X and which has the property that for every harmonic function h on a strictly affinoid domain U of X with $u|_{\partial U} \leq h|_{\partial U}$, we have $u|_U \leq h$.

Proposition 5.1. *Let X be a smooth, strictly k -analytic curve and let $u: X \rightarrow \mathbf{R}$ be an $\mathbf{R}$ -PL function. Then u is subharmonic at $x \in X$ in the sense of Thuillier if and only if $L_u(x)$ is nef.*

Proof. First we treat the case when k is algebraically closed. As explained in 2.5, there exists a compact strictly analytic neighborhood U of x in X , a formal k° -model $\mathfrak{U}$ of U , and a formal model $\mathcal{L}$ of $\mathcal{O}_U$ such that $u|_U = h_{\mathcal{L}}$. Since k is algebraically closed, we may assume that the special fiber $\mathfrak{U}_s$ is reduced, which implies that $\mathfrak{U}$ is covered by formal affines of the form $\text{Spf}(\mathcal{A}^\circ)$ for a strictly k -affinoid algebra $\mathcal{A}$ [GJR21, §2.4]. It follows from [Ber90, Proposition 2.4.4] that for every irreducible component Y of $\mathfrak{U}_s$, there is a unique point $\xi_Y \in U$ reducing to the generic point of Y . In the formalism of real valued forms and currents on Berkovich spaces [CD12], it is well known that $d'd''[u]|_U$ is the discrete measure on U supported on the points ξ_Y of U for Y an irreducible component of $\mathfrak{U}_s$, with multiplicity at ξ_Y equal to $\deg_s(Y)$. See [CD12, §6.9] for a more general statement. Wanner showed that subharmonicity of u on U is equivalent to the positivity of the current $d'd''[u]$: see [Wan19, Theorems 1 and 2] and use the fact that X is locally algebraic [vdP80]. Thus $\mathcal{L}_s$ is nef if and only if u is subharmonic on U . Suppose that u is subharmonic at x . Shrinking U , we may assume that u is subharmonic on U . This implies that $\mathcal{L}_s$ is nef, so it follows from [GJR21, Proposition 7.1] that $L_u(x)$ is nef. Conversely, if $L_u(x)$ is nef, then by [GJR21, Proposition 7.1], we may shrink U so that $\mathcal{L}_s$ is nef, which as above implies that u is subharmonic on U .

It remains to show that subharmonicity of u at x and numerical effectivity of $L_u(x)$ can be checked after extending scalars to the completion of an algebraic closure $K = (\bar{k})^\wedge$. By [GJR21, Proposition 7.6(2)], one can check that $L_h(x)$ is nef after base change to K . The same is true for subharmonicity: this follows directly from Thuillier’s definition and the fact that the base change of a harmonic function remains harmonic [Thu05, Proposition 2.3.18]. $\square$

Our next goal is a characterisation of semipositive $\mathbf{R}$ -PL functions by restriction to subcurves as in the complex case. We need the following characterization of boundary points: